Stress and Telomere Biology: A Lifespan Perspective

In the past decade, the growing field of telomere science has opened exciting new avenues for understanding the cellular and molecular substrates of stress and stress-related aging processes ver the lifespan. Shorter telomere length is associated with advancing chronological age and also increased disease morbidity and mortality. Emerging studies suggest that stress accelerates the erosion of telomeres from very early in life and possibly even influences the initial (newborn) setting of telomere length. In this review, we highlight recent empirical evidence linking stress and mental illnesses at various times across the lifespan with telomere erosion. We first present findings in the developmental programming of telomere biology linking prenatal stress to newborn and adult telomere length. We then present findings linking exposure to childhood trauma and to certain mental disorders with telomere shortening. Last, we review studies that characterize the relationship between related health-risk behaviors with telomere shortening over the lifespan, and how this process may further buffer the negative effects of stress on telomeres. A better understanding of the mechanisms that govern and regulate telomere biology throughout the lifespan may inform our understanding of etiology and the long-term consequences of stress and mental illnesses on aging processes in diverse populations and settings.